SHUANG / A RETAINED LISTENING STUDY

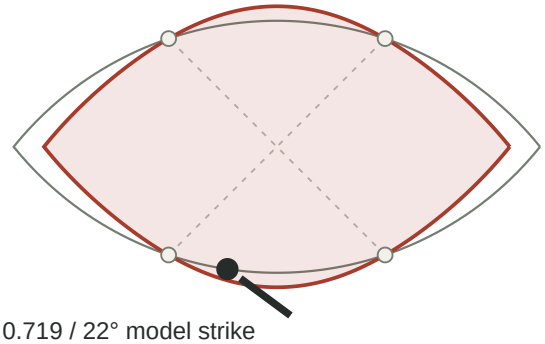
Two voices cast into one bell

Selected: **No. 8**. Strike 22°. Published frequencies. Restore this bench.

In an ideal round, symmetric bell, a rotated pair of bending patterns can share one frequency. A noncircular shape can separate that pair. Changing where such a bell is struck can change which patterns are excited. It does not retune the metal. This study isolates two published resonances, then compares measured dimensions with two readings of a transmitted Chinese proportion rule.

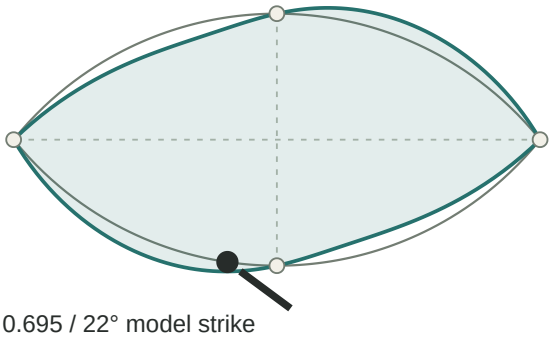
ONE RIM / TWO IDEAL BENDING PATTERNS

a / lower resonance



coupling 0.719 / 22° model strike

b / upper resonance



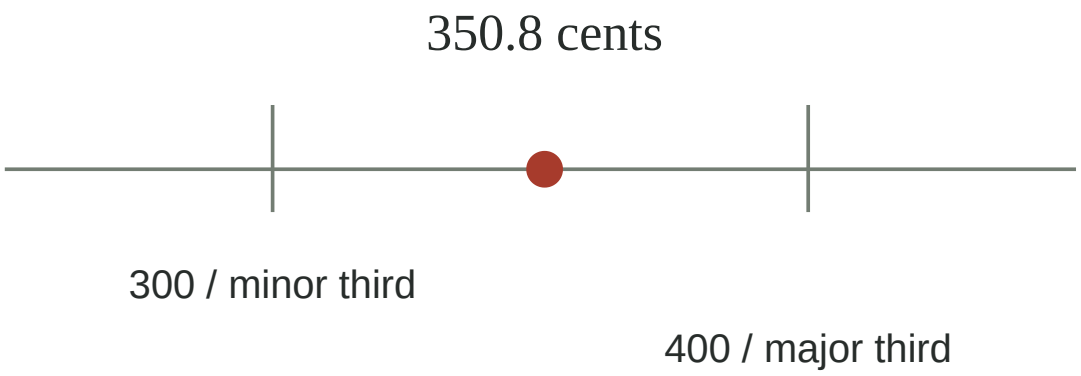
coupling 0.695 / 22° model strike

Open circles: no radial motion in that ideal pattern.

New quadrupole analogy on a lens-like rim. Neither a scanned mode shape nor a prediction of a particular ancient bell. Couplings are model coefficients, not measured energies. Drawing deformation is enlarged.

1,214.1 Hz / 1,486.8 Hz

The current interval is 350.788 cents.



The interval diagram always uses reported Hz. The optional projection rounds each resonance to a 100-cent grid referenced to 364.07 Hz; it does not establish ancient equal temperament or use A = 440 Hz.

Keep the sound and its limits together

Each embedded WAV is a newly generated two-second mono 44.1 kHz, 16-bit PCM signal: two sine waves, an 8 ms attack and a 0.65 s exponential decay, ending with a 25 ms fade. The amplitude weights are cos(2θ) and sin(2θ). Higher modes, impact noise, actual damping, room sound and the original timbre are absent. These are not recordings of the ancient objects.

a alone: face end of the model

Keep this WAV

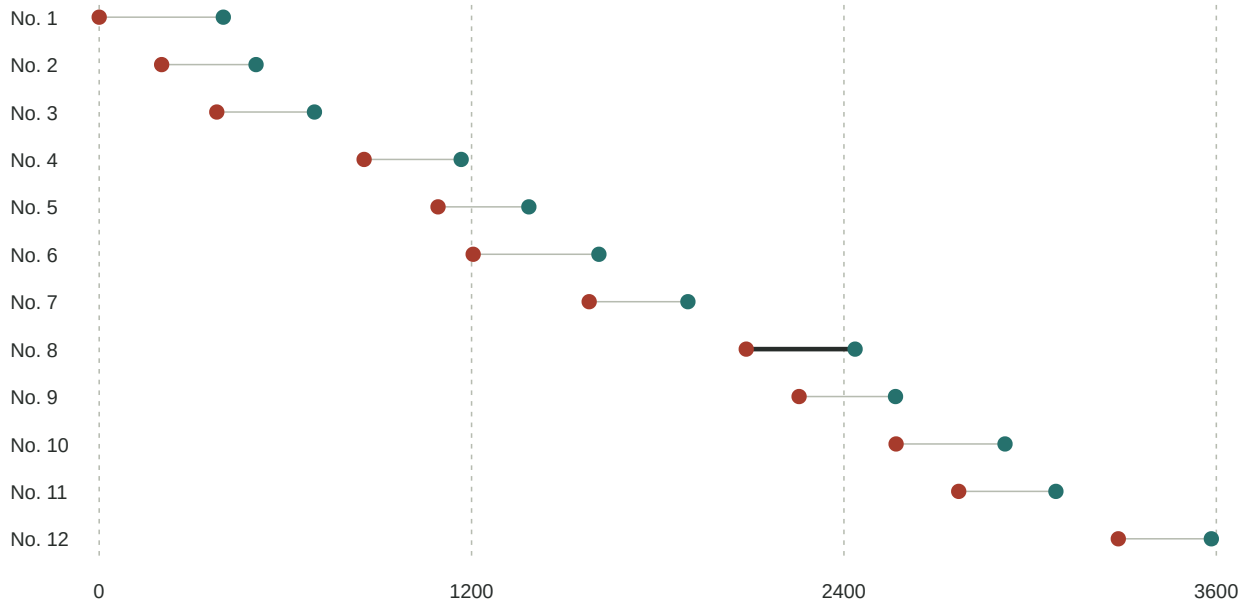
b alone: flank end of the model

Keep this WAV

Selected mixture: 22°

Keep this WAV

Twenty-four resonances, fourteen sets of dimensions



Pitch map: reported frequencies; horizontal axis in cents above 364.07 Hz. Red = a, teal = b. No. 8 and No. 10 are near 351 cents, between modern 300-cent and 400-cent thirds. Ancient tuning and corrosion cannot be inferred from a rounding grid.

Bell	ridge length / cm	body length / cm	tip-to-tip width / cm	mouth depth / cm	crown length / cm	crown breadth / cm	Mass / g	a / Hz	b / Hz	a / cents[1]	b / cents[1]
No. 1	22.3	18.0	17.7	14.1	14.2	11.3	4578	364.07	458.70	0	400
No. 2	21.0	16.8	16.8	13.3	13.3	10.5	3622	408.95	487.51	201	506
No. 3	19.9	15.8	15.7	12.4	12.7	9.7	3048	453.15	543.50	379	694
ROM 1	17.9	14.5	14.0	11.4	11.5	9.1	not reported	not reported	not reported	not reported	not reported
No. 4	17.0	13.6	13.5	10.6	11.0	8.5	2025	596.14	714.18	854	1167
No. 5	16.0	12.7	12.5	10.0	10.3	8.2	1760	684.07	810.09	1092	1385
No. 6	15.3	12.3	12.1	9.8	10.1	8.0	1623	730.30	922.91	1205	1610
ROM 2	14.5	11.9	11.4	9.0	9.4	7.4	not reported	not reported	not reported	not reported	not reported
No. 7	13.7	11.0	11.1	8.9	9.2	7.2	1363	906.32	1089.2	1579	1897
No. 8	12.0	9.8	10.0	7.8	7.9	6.2	935	1214.1	1486.8	2085	2436
No. 9	11.3	9.2	9.2	7.4	7.4	5.8	815	1339.6	1603.0	2255	2566
No. 10	10.3	8.4	8.4	6.8	6.7	5.2	670	1604.7	1965.4	2568	2919
No. 11	9.9	8.1	8.1	6.4	6.6	4.9	605	1803.1	2160.7	2770	3083
No. 12	8.7	7.2	7.1	5.7	5.9	4.5	510	2426.8	2885.7	3284	3584

[1] The paper's printed relative cents, referenced to No. 1 a = 364.07 Hz. All frequencies are as reported for 30°C. Printed decimals are retained, not an uncertainty estimate. Table 1 supplies dimensions for 14 bells; Table 2 supplies two resonances for the 12 Kyoto bells. ROM means Royal Ontario Museum. Empty CSV cells mean unreported, never zero.

Keep all measurements as CSV

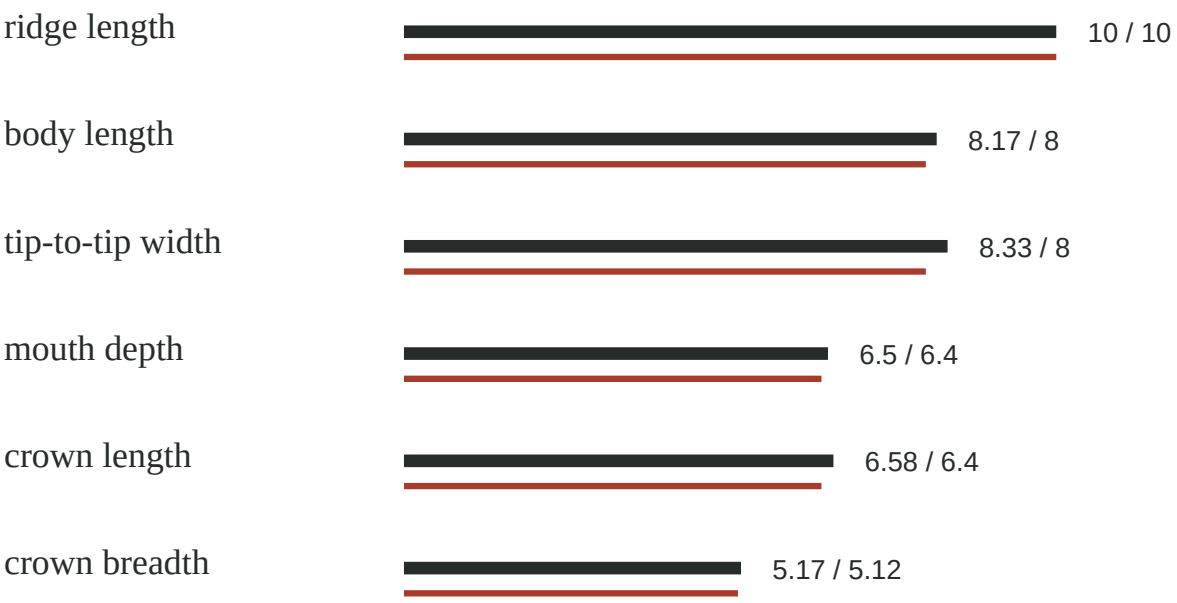
Intervals in this workshop are computed from the reported Hz, rather than subtracting independently rounded printed relative cents. For example, No. 2 has printed values 201 and 506, while its interval calculated from Hz rounds to 304 cents.

Read the bell-maker's proportions

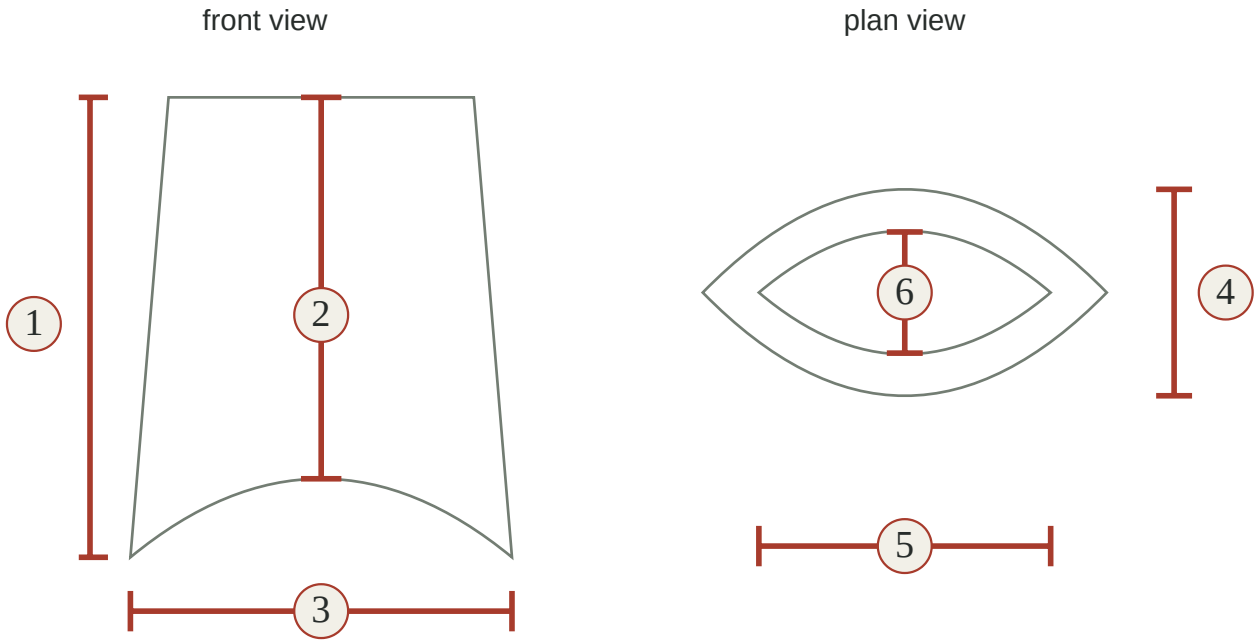
Source: Zhouli, Kaogongji, Fu craft section; Wikisource revision 2159477. The selections retain its characters, with editorial full stops at shortened boundaries, new English explanations and modern Mandarin pinyin. Pinyin is not reconstructed ancient pronunciation.

The selected interpretation gives 10 : 8 : 8 : 6.4 : 6.4 : 5.12, in the order used below. Black bars show the selected bell, normalized to a ridge length of ten; red bars show the selected reading. Agreement is not evidence that a specific object was made from this prescription.

NORMALIZED TO A RIDGE LENGTH OF TEN



LOCATE THE DIMENSIONS / SCHEMATIC, NOT TO SCALE



1 ridge · 2 body · 3 tip width · 4 mouth depth · 5 crown length · 6 crown breadth

READING 1 / The bell-makers

鳧氏為鐘。

The Fu craftspeople make bells.

鐘 / zhōng / bell

This is a craft heading in the transmitted Kaogongji, within the Zhouli. It names a class of work, not a surviving signed measurement sheet for the Piao-shi set. The text and the archaeological objects are different kinds of evidence.

READING 2 / Name the two tips

兩欒謂之銑，銑間謂之于。

The two projecting corners are called xian; the space between the xian is called yu.

銑間 / xiǎn jiān / the interval between the tips

間 names an interval or intervening space. The measured tip-to-tip width, xianjian, is the representative size used in the published frequency fit. It is not the full bell height and not its wall thickness.

READING 3 / A vocabulary of regions

于上謂之鼓，鼓上謂之鉦，鉦上謂之舞。

Above the mouth region is what is called gu; above gu is zheng; above zheng is wu.

舞 / wǔ / the crown or top plate here

A familiar character can have a technical meaning. 舞 names the top plate in this description. The six measured dimensions use these part names; they do not all measure the same direction. The diagram is a new schematic, not a measured contour of an individual bell.

READING 4 / Start with ten

十分其銑，去二以為鉦，以其鉦為之銑間。

Divide the ridge length into ten parts; remove two to make the body length. Use that body length for the distance between the tips.

十分 … 去二 / shí fēn … qù èr / divide into ten … remove two

Both interpretations begin with 10 : 8 : 8 for ridge length, body length and tip-to-tip width. The later subtractions create the disagreement. The English explains the operation rather than pretending the text supplies centimetres.

READING 5 / Two parts of which whole?

去二分以為之鼓間。

Remove two parts to make the distance across the gu region.

分 / fēn / a part or division

If the two parts still belong to the original ten, eight becomes six. If the new dimension is divided into ten again, eight becomes 6.4. The two readings therefore make different measurable predictions. The paper discusses both interpretations; this control does not establish one as the unique ancient practice.

READING 6 / Measure the crown

以其鼓間為之舞修，去二分以為舞廣。

Use that gu interval for the crown's long dimension; remove two parts to make its breadth.

廣 / guǎng / breadth

The fixed-part reading gives a crown 6 by 4, while successive four-fifths reductions give 6.4 by 5.12. Compare both with surviving dimensions. 修 is the transmitted form used here; the 1989 paper prints the variant 脩. Modern Mandarin pinyin is a reading aid, not a reconstruction of ancient pronunciation.

Size is not enough

The 1989 paper fits frequencies in this set approximately as inverse tip-to-tip width squared, and relates this to approximately constant wall thickness inferred from mass. It is an empirical relation within this set, not a universal law of bronze bells.

A conditional thin-shell comparison, $f \propto h/L^2$, gives different answers depending on what grows. At scale $\lambda = 2$, uniformly scaling the bell and its wall gives f/λ, while keeping wall thickness fixed gives f/λ^2. For this selected a resonance, that is 607.05 Hz or 303.52 Hz. Alloy, profile and all other shape details are assumed unchanged. This is a scaling exercise, not a casting simulator.

A surviving object, separately identified



Cleveland Museum of Art, Bell (Bo Zhong), 1962.44; early 400s BCE; John L. Severance Fund. Museum Open Access photograph and object data: CC0.

A separate ancient bell, not one of the measured Piao-shi bells. Its photograph illustrates the material object; no sound of this museum object is claimed. Museum object record; Open Access rights.

Retained sources and rights

Junji Takahashi (1989), “Size-frequency relation and tonal system in a set of ancient Chinese bells: Piao-shi bianzhong,” Journal of the Acoustical Society of Japan (E) 10(5), 299–303. DOI: 10.1250/ast.10.299. Official article. Tables 1 and 2 are transcribed as facts. The copyrighted paper and its page images are not embedded.

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- cleveland-1962.44.jpg
- kaogongji-fu-revision-2159477.txt
- kaogongji-selections.txt
- measurements.json
- provenance.json

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SHUANG: SOURCE, IMAGE AND FONT NOTICES

New code and original schematic diagrams: MIT.
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New explanatory teaching text, English readings and the adapted reader:
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MEASURED FACTS
Junji Takahashi (1989), "Size-frequency relation and tonal system in a set of
ancient Chinese bells: Piao-shi bianzhong", Journal of the Acoustical Society
of Japan (E) 10(5), 299-303. DOI 10.1250/ast.10.299.
https://www.jstage.jst.go.jp/article/ast1980/10/5/10_5_299/_article
Dimensions and mass: Table 1, printed p. 300. Resonances and printed relative
pitches: Table 2, p. 301, as reported for 30 degrees Celsius.
The numerical data are transcribed facts. The copyrighted paper, its page
images and its figures are not redistributed here. The ROM objects have
dimensions but no reported frequencies or mass in these tables.

CHINESE TEXT
Zhouli, Dongguan Kaogongji, Fu craft section. Ancient text: public domain.
Digital reading credited to Chinese Wikisource contributors, revision 2159477,
2022-06-29. Selected characters and internal punctuation retained; intervening text omitted.
Full stops at truncated excerpt boundaries are editorial additions. The complete
Fu section is retained separately with its original punctuation.
https://zh.wikisource.org/w/index.php?oldid=2159477&title=周禮/冬官考工記
Digital selection and adapted reader: CC BY-SA 4.0. English and teaching notes
are new, not a historical translation. Modern Mandarin pinyin is not ancient
pronunciation. The character 修 follows the cited digital reading; the 1989
article prints 脩. Neither reading is silently replaced with the other.

PHOTOGRAPH
Bell (Bo Zhong), early 400s BCE, China, bronze; Cleveland Museum of Art 1962.44.
John L. Severance Fund.
https://www.clevelandart.org/art/1962.44
Museum Open Access photograph and object data: CC0.
https://www.clevelandart.org/open-access
The 463 x 600 JPEG is retained without retouching. Its identity is recorded in
provenance.json. This is a separate object, not a Piao-shi measurement-table bell.
No sound recording or measured pitch of the Cleveland object is claimed.

SYNTHESIZED SOUND AND NEW SCHEMATICS
Each two-second sound is generated from the selected two frequencies and a
modern, explicitly chosen decay envelope. It omits actual higher modes, impact
noise, room acoustics and measured damping. It is not a museum recording or a
reconstruction of ancient timbre. The lens-like rim and quadrupole drawings are
new teaching analogies, not measured or finite-element mode shapes.
The optional modern semitone grid is referenced to 364.07 Hz. It does not
establish ancient equal temperament or an A = 440 Hz reference.

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